

PERSONAL PROJECT REPORT

# VK DSA Sheet

*Master Patterns. Solve Problems. Level Up.*

*A free, open-source, pattern-based DSA practice platform unifying LeetCode, GeeksforGeeks, TakeUForward and Codeforces — with built-in progress tracking, revision management, and a personal knowledge base.*

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## 1. Introduction & Problem Statement

Every coding platform promotes its own problem set — LeetCode, TakeUForward, GeeksforGeeks and Codeforces each have strengths, but none give students one unified, pattern-based place to practice, track, and revise.

| Platform      | What They Offer      | What's Missing                                 |
|---------------|----------------------|------------------------------------------------|
| LeetCode      | Huge question bank   | No pattern roadmap, no cross-platform tracking |
| TakeUForward  | Striver's SDE Sheet  | Limited to their own curated list              |
| GeeksforGeeks | Topic-wise problems  | No unified progress tracking across platforms  |
| Codeforces    | Competitive problems | No structured DSA sheet                        |

### The Real Issue

- Students jump between four or five platforms with no single place to track everything
- Sheets exist, but they are just lists — no pattern-based learning
- No way to mark revision status separately from completion
- No personal bookshelf, link storage, or notes alongside practice
- No calendar view to visualize coding consistency
- No difficulty-wise breakdown (Easy / Medium / Hard) with progress

*There is no completely free website that combines LeetCode + GFG + TUF + Codeforces + Pattern Names + Progress Tracker all in one place — until now.*

## 2. The Idea Behind VK DSA Sheet

The frustration was simple: every creator and every platform promotes their own sheet. As a student, the goal was one place to see all patterns — not just topics like "Arrays," but actual patterns like Sliding Window, Two Pointers, and Kadane's Algorithm — pull questions from every major platform, and track progress, revision, and personal notes together.

### What the Platform Set Out to Deliver

- See ALL patterns, not just broad topics
- Pull questions from LeetCode, GFG, TUF and Codeforces — all in one place
- Track what has been COMPLETED
- Track what needs to be REVISED, separately from completion
- Store favourite BOOKS
- Save important LINKS
- Write personal NOTES
- See practice history on a CALENDAR
- See a difficulty breakdown — Easy, Medium, Hard
- Have all of this PERSIST with a secure login

*"No such website existed. So I built one."*

### 3. Core Features

#### Pattern-Based Learning

Instead of solving questions at random, learners work through the way interviewers actually think — by recognizing patterns, not memorizing individual problems.

- 16 pattern categories: Arrays, Binary Search, Strings, Bit Manipulation, Linked List, Two Pointers, Sliding Window, Stacks & Queues, Greedy Algorithms, Heaps, Binary Trees, BST, Graphs, Tries, Dynamic Programming and more
- Sub-pattern breakdown for each category — e.g. Arrays splits into Array Traversal, Rotate Array, Brute Force Subarray, Kadane's Algorithm, and Dutch National Flag
- Every pattern sources problems from LeetCode, TakeUForward, GeeksforGeeks and Codeforces

#### Personal Knowledge Base

| Feature | Description                                                 |
|---------|-------------------------------------------------------------|
| Search  | Quickly find any pattern or question                        |
| Books   | Store favourite DSA books plus recommended titles           |
| Links   | Save important links — YouTube videos, articles, editorials |
| Notes   | Write personal notes for any topic or question              |

#### User Authentication

- Secure Login / Sign-Up powered by Firebase Authentication
- Persistent data — progress, books, links, and notes all saved to the account
- Cross-device sync — log in from anywhere and pick up where you left off

## 4. Progress Tracking System

Every question can be marked complete AND separately flagged for revision — a dual-tracking feature missing from almost every existing DSA sheet.

| Element                | Description                                                    |
|------------------------|----------------------------------------------------------------|
| Completion Tick        | Mark a question as done                                        |
| Revision Tick          | Separately mark a question for a second pass                   |
| Progress Bars          | Visual progress for each sub-pattern, e.g. 5 / 16              |
| Circular Progress Ring | Overall completion indicator, e.g. 11 / 65                     |
| Difficulty Counter     | Tracks Easy, Medium and Hard problems solved                   |
| Beats Percentage       | Shows how a learner compares against others, e.g. "Beats 2.5%" |

### Example — Dashboard Snapshot

| Metric               | Value            |
|----------------------|------------------|
| Total Solved         | 12 Problems      |
| Easy / Medium / Hard | 5 / 7 / 0        |
| Overall Completion   | 11 / 65 patterns |
| Beats Percentage     | 2.5%             |

## 5. Practice Calendar & Knowledge Base

### GitHub-Style Practice Calendar

- A GitHub-style calendar shows which days a learner practiced
- Daily streak tracking highlights active days
- Monthly view lets learners navigate through their history
- A detailed practice log records questions solved per day

### Personal Knowledge Base (Sidebar)

Beyond the sheet itself, the platform doubles as a coding-journey companion — not just a problem list.

| Feature | Purpose                                          |
|---------|--------------------------------------------------|
| Books   | Recommended DSA books plus user-added favourites |
| Links   | Saved YouTube videos, blog posts and editorials  |
| Notes   | Personal notes written for any topic or problem  |
| Search  | Instant lookup across every pattern and question |

## 6. Technology Stack

| Layer              | Technology                                |
|--------------------|-------------------------------------------|
| Frontend           | React, JavaScript, HTML5, CSS3            |
| Backend / Database | Firebase Authentication + Cloud Firestore |
| Deployment         | Vercel                                    |
| Version Control    | Git & GitHub                              |

### Libraries & Tools Used

| Library / Tool    | Purpose                                      |
|-------------------|----------------------------------------------|
| React             | Component-based UI framework                 |
| Firebase Auth     | User authentication (Login / Sign-Up)        |
| Cloud Firestore   | Real-time database for storing user progress |
| React Router      | Client-side routing and navigation           |
| CSS3 / Custom CSS | Styling with dark theme and animations       |
| Vercel            | Production deployment and hosting            |
| Git & GitHub      | Version control and collaboration            |



## 7. System Architecture & User Flow

The application takes a visitor from an anonymous browse session all the way to a synced, cross-device practice habit.

| Step                 | What Happens                                                 |
|----------------------|--------------------------------------------------------------|
| 1. Visit & Browse    | Explore all DSA patterns — no login required                 |
| 2. Login / Sign Up   | Secured via Firebase Authentication                          |
| 3. Pick a Pattern    | Pattern -> Sub-pattern -> Question                           |
| 4. Solve & Mark      | Mark complete, or flag for revision                          |
| 5. Sync in Real Time | Progress saved instantly to Cloud Firestore                  |
| 6. Track Consistency | Calendar, stats, notes and links — available on every device |

### Flow Summary

*User Visits -> Browses Patterns -> Login / Sign Up -> Selects Pattern -> Solves Question -> Marks Complete or Revision -> Progress Saved to Firebase -> Views Calendar & Stats -> Stores Books / Links / Notes -> Continues from Any Device*

## 8. DSA Patterns Covered

Sixteen pattern categories give a complete, interview-ready roadmap across every major data structure and algorithm family.

| #  | Pattern             | Sub-Patterns                                     |
|----|---------------------|--------------------------------------------------|
| 01 | Arrays              | Traversal, Rotate, Kadane's, Dutch National Flag |
| 02 | Binary Search       | Standard, on Answers, on Rotated Arrays          |
| 03 | Strings             | Matching, Palindromes, Anagrams                  |
| 04 | Bit Manipulation    | XOR Tricks, Bit Masking, Power of Two            |
| 05 | Linked List         | Reversal, Fast & Slow Pointer, Merge             |
| 06 | Two Pointers        | Opposite / Same Direction, Three Pointers        |
| 07 | Sliding Window      | Fixed, Variable, Shrinkable                      |
| 08 | Stacks & Queues     | Monotonic Stack, Next Greater Element            |
| 09 | Greedy Algorithms   | Activity Selection, Fractional Knapsack          |
| 10 | Heaps               | Top-K Elements, Merge K Sorted, Median           |
| 11 | Binary Trees        | Traversals, Views, Diameter, LCA                 |
| 12 | BST                 | Search, Insert, Delete, Validate                 |
| 13 | Graphs              | BFS, DFS, Dijkstra, Topological Sort             |
| 14 | Tries               | Insert, Search, Prefix Matching                  |
| 15 | Dynamic Programming | 1D / 2D DP, Knapsack, LIS, LCS                   |
| 16 | More...             | Additional patterns added regularly              |

## 9. Comparative Analysis

VK DSA Sheet is the only platform in this comparison that combines a pattern-wise roadmap, multi-platform questions, dual progress tracking, a practice calendar, and a personal knowledge base — while remaining completely free.

| Feature                        | LeetCode | TUF Sheet | GFG Sheet | NeetCode | VK DSA Sheet |
|--------------------------------|----------|-----------|-----------|----------|--------------|
| Pattern-wise Roadmap           | —        | ~         | —         | ~        | ✓            |
| Multi-Platform Questions       | —        | —         | —         | —        | ✓            |
| Progress Tracking              | ~        | ✓         | —         | ✓        | ✓            |
| Revision Tracking              | —        | —         | —         | —        | ✓            |
| Practice Calendar              | ✓        | —         | —         | —        | ✓            |
| Personal Notes / Books / Links | —        | —         | —         | —        | ✓            |
| Completely Free                | —        | ✓         | ✓         | —        | ✓            |

Legend: ✓ = fully supported   ~ = partially supported   — = not supported

## 10. Product Walkthrough

The dashboard is the first thing a learner sees after logging in — pattern categories, a live progress ring, difficulty breakdown, and a GitHub-style calendar, all visible at a glance.



Fig. 1 — All DSA Patterns dashboard, showing live progress and calendar tracking

## 11. Design Philosophy & Roadmap

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### Design Principles

- Dark theme — easy on the eyes for late-night coding sessions
- Purple & cyan gradient — a modern, developer-friendly identity
- Fully responsive — works cleanly across desktop, tablet and mobile
- Real-time sync — Firebase keeps every device instantly up to date

### Roadmap Ahead

- Expand beyond 16 patterns with community-suggested additions
- Add contest / company-wise question tags (FAANG, product-based)
- Introduce streak rewards and a leaderboard for friendly competition
- Build a mobile app version for on-the-go revision

## 12. Conclusion

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VK DSA Sheet was built to solve a problem every DSA learner faces: too many platforms, no single place to track patterns, progress, and revision. By unifying LeetCode, GeeksforGeeks, TakeUForward and Codeforces into one pattern-based roadmap — with real-time tracking, a practice calendar, and a personal knowledge vault — it gives learners one home for their entire coding-interview journey, completely free.

Try it now: [vk-dsa-hub.netlify.app](https://vk-dsa-hub.netlify.app)